

“ELECTRO PONG”

Game Design Document

Concept: A skill- and strategy-based game.

Core Gameplay

- The ball bounces back and forth across the arena.
- Players have inertia and weight, creating a movement feel similar to playing tennis.
- Players can choose the angle of their return, directing the ball to the left, center, or right.

Features

Energy System

- The player gains Energy Points by successfully performing a mechanic similar to *League of Legends*' last-hit system, possibly by hitting the ball exactly at the center of the paddle/player.
- Once the Energy Bar is fully charged, the player automatically receives a random power from a finite pool of available powers.
- There are both offensive and defensive powers.
- The player receives clear visual feedback indicating which power is currently available.
- Offensive powers can only be activated while the ball is in the player's own quarter of the field.
- Defensive powers can only be activated while the ball is in the player's own half of the field.
- If the player's Energy Bar is fully charged and they have not yet used their current power, the bar remains full. As a result, no additional Energy Points can be earned until that power is used.
- The ball gradually accelerates as the rally continues, up to a maximum speed determined by the selected difficulty.
- The player's movement speed also increases throughout the rally, but at a slower rate than the ball.
- Once per rally, the player may discard their current power (or empty their Energy Bar) without receiving anything in return.
- Matches are played in sets. Each set consists of games, and each game is won by reaching a certain number of goals. This system is directly inspired by tennis.
- Trophy system as proof of defeating the AI.

- Defeating the AI on Hard difficulty unlocks the new **Expert** difficulty as well as skin customization.

Offensive Powers

Ball Acceleration

The player infuses the ball with stored energy, granting it a significant speed boost that can catch the opponent off guard. The ball remains accelerated until the opponent successfully hits it back or a goal is scored.

If the opponent successfully returns the ball, they immediately gain a random **50–60%** Energy charge. If they already have a charged power, this Energy is granted after they use their current power. This bonus cannot be stacked.

Opponent Slowdown

The player emits an electromagnetic pulse that slows the opponent until they successfully return the ball or a goal is scored.

If the opponent returns the ball, they receive a random **50–60%** Energy charge. If they already have a charged power, this Energy is granted after they use it. This effect cannot be stacked.

Steal Opponent's Power

The player removes the opponent's currently charged power.

As compensation, the opponent immediately gains a random **70–80%** Energy charge. If the opponent does not yet have a fully charged power, the player instead steals a percentage of their current Energy charge.

If the opponent was affected by **Ball Acceleration**, they lose that power and immediately receive the compensation Energy described above.

Defensive Powers

Player Acceleration

The player converts their stored energy into a temporary movement speed boost that lasts until they hit the ball or concede a goal.

If the player was slowed by **Opponent Slowdown**, this power removes the slowdown and restores normal movement speed.

If the player was affected by **Ball Acceleration**, they receive the **Player Acceleration** effect and immediately gain the corresponding Energy compensation.

Extended Reach

The player projects an energy beam upward and downward, temporarily extending their paddle to better cover the goal.

This power requires precise timing, as it only lasts for a fraction of a second after activation.

If the player was affected by **Ball Acceleration**, they receive the **Extended Reach** effect and immediately gain the corresponding Energy compensation.

Power Swap

The player randomly receives any available power except **Power Swap** itself.

Base Ability System

Both players have three core abilities. Each ability can be countered through another ability, quick reflexes, or accurate prediction.

Core Abilities

- Fast Ball
- Slow Ball
- Multi Ball

Corresponding Counters

- Time Slow
- Magnetism Cancel
- Paddle Extension / Bilateral Energy Beam